

PROJECT REPORT

WATER LEVEL INDICATOR WITH PCB DESIGN

1. Title

Design and Implementation of Water Level Indicator using PCB

2. Objective

The objective of this project is to design a simple and low-cost water level indicator circuit and convert it into a PCB layout. The system indicates different water levels in a tank using LEDs and/or buzzer indication.

3. Abstract

Water overflow and water wastage are common problems in homes, industries, and agriculture. This project presents a water level indicator circuit designed using electronic components and implemented on a PCB. Conductive probes are placed at different levels inside the water tank. When water touches each probe, the corresponding circuit path is completed, and the related LED glows to indicate the water level.

The circuit is first designed in schematic form and then converted into a PCB layout using Altium Designer. The PCB design improves reliability, reduces wiring complexity, and makes the circuit compact and professional.

4. Components Used

S.No	Component	Purpose
1	Resistors	Current limiting and biasing
2	LEDs	Indicate different water levels
3	Transistors	Act as switching devices
4	Buzzer	Gives alert when tank is full
5	Connector/Header	Connects water sensing probes
6	Power Supply	Provides DC voltage to the circuit
7	PCB Board	Holds and connects all components

5. Working Principle

The water level indicator works based on the conductivity of water. Water can conduct a small amount of electric current. Probes are placed at different levels in the tank such as low, medium, high, and full level.

When the water reaches a particular probe, the circuit path is completed through the water. This signal activates the corresponding transistor or indicator section. As a result, the LED connected to that level turns ON. When the water reaches the full-level probe, the buzzer also turns ON to alert the user.

6. Block Diagram



7. Circuit Description

The circuit consists of multiple sensing points connected to different water levels. Each sensing point is connected to a switching circuit. The switching circuit may use transistors to drive LEDs and buzzer outputs.

Each LED represents a particular water level. For example:

Water Level	Indication
Low Level	LED 1 ON
Medium Level	LED 2 ON
High Level	LED 3 ON
Full Level	LED 4 ON and buzzer ON

When the water touches the common probe and any level probe, the corresponding transistor gets activated. The transistor acts as a switch and allows current to flow through the LED, making it glow.

8. PCB Design Procedure

The PCB was designed using **Altium Designer**. The following steps were followed:

1. Created a new PCB project.
2. Designed the schematic circuit.

3. Selected proper components and footprints.
4. Compiled the schematic and checked for errors.
5. Updated the schematic design to PCB.
6. Placed the components properly on the PCB board.
7. Routed the tracks between components.
8. Added silkscreen names and board outline.
9. Performed Design Rule Check.
10. Generated output files such as schematic, PCB layout, Gerber files, NC drill files, and BOM.

9. PCB Design Considerations

The following points were considered during PCB design:

- Proper component placement was done to reduce routing complexity.
- Track width was selected based on current requirement.
- Sufficient clearance was maintained between tracks.
- Silkscreen labels were added for easy component identification.
- Connectors were provided for water probes and power supply.
- Design Rule Check was performed to ensure the PCB had no major errors.

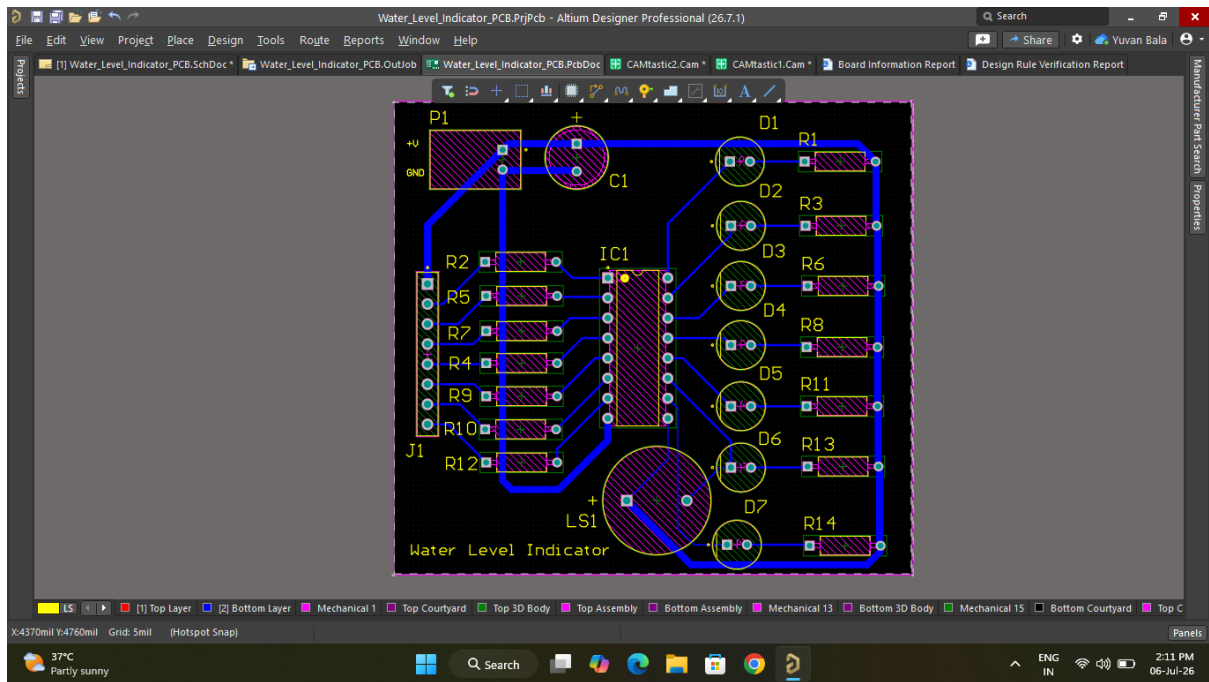
10. Testing Procedure

After completing the PCB design, the circuit can be tested using the following method:

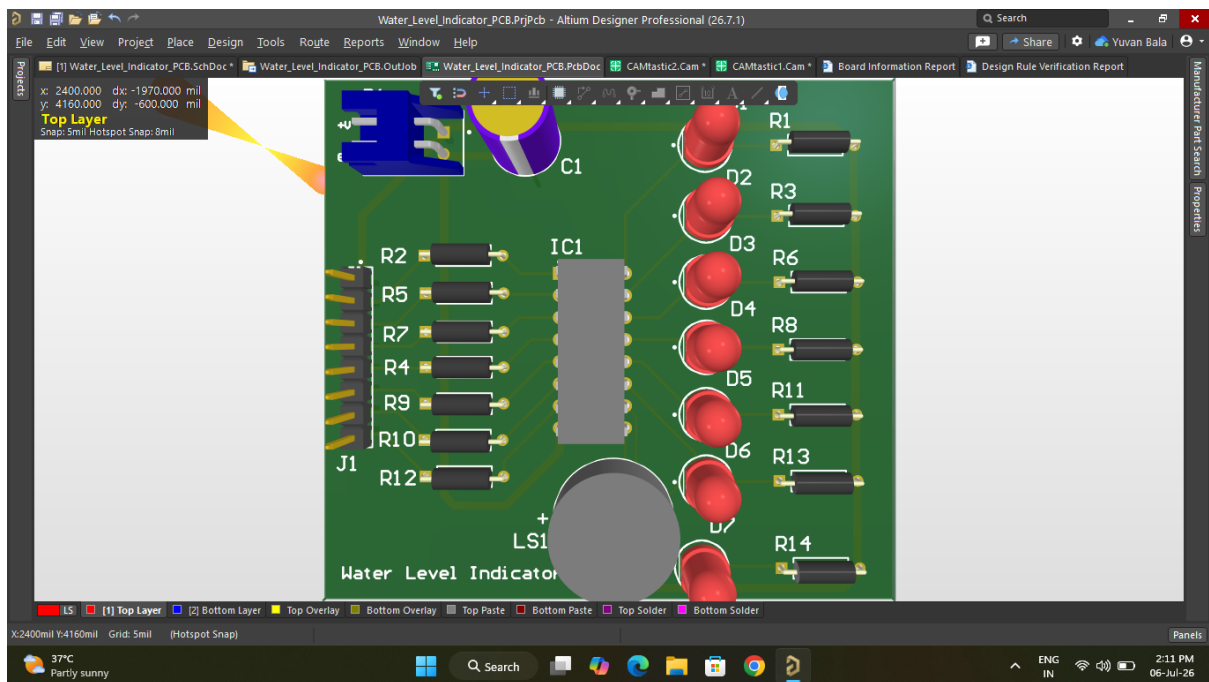
1. Connect the required DC power supply.
2. Connect wires from the probe connector to different water levels.
3. Dip the common probe and level probes into water step by step.
4. Observe the LEDs turning ON according to the water level.
5. Check whether the buzzer turns ON at full level.
6. Verify that all indicators work correctly.

11. Result

The water level indicator circuit was successfully designed and converted into a PCB layout using Altium Designer. The design helps to indicate water levels clearly using LED indicators and buzzer alert. The PCB design makes the circuit compact, reliable, and suitable for practical use.



Top View



3D View

12. Applications

- Home water tanks
- Overhead tanks
- Industrial water storage systems
- Agriculture water tanks

- Small automation projects
- Student electronics projects

13. Advantages

- Simple and low-cost design
- Easy to understand and implement
- Reduces water wastage
- Compact PCB-based circuit
- Useful for practical applications

14. Limitations

- Manual probe cleaning may be required due to corrosion or dust.
- The circuit only indicates water level; it does not automatically control the motor unless a relay/motor control circuit is added.
- Conductivity may vary depending on water quality.

15. Future Scope

- Automatic motor ON/OFF control can be added.
- Microcontroller-based digital display can be included.
- Wireless water level monitoring can be implemented.
- IoT-based mobile notification system can be added.
- Non-contact sensors can be used to avoid probe corrosion.

16. Conclusion

This project successfully demonstrates the design of a water level indicator circuit and its implementation as a PCB. The project provides practical knowledge of circuit design, component selection, schematic creation, PCB layout, routing, and design rule checking. It is a useful beginner-level electronics and PCB design project with real-time applications.

17. Files Included in Project Folder

- Schematic file
- PCB layout file
- Gerber files
- NC drill files
- Bill of Materials

- 3D view screenshot
- Schematic screenshot
- PCB layout screenshot
- Project report
- README file

18. Software Used

Altium Designer – Schematic design, PCB layout, routing, DRC checking, and output file generation.